

CHERBURG MAUPERTUS, France

WMO: 070240

Lat: 49.6528N

Lon: 1.4636W

Elev: 140

StdP: 99.66

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.0	0.1	-5.0	2.5	0.8	-3.6	2.8	1.4	14.5	8.4	13.3	8.0	3.9	160	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	24.0	18.3	22.1	17.1	20.7	16.5	18.9	22.5	18.0	20.9	17.3	19.7	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.8	20.4	16.8	12.2	19.3	16.2	11.7	18.5	54.3	22.5	51.3	21.0	49.0	19.8	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	10.9	9.7	-3.0	29.4	1.6	2.3	-4.2	31.0	-5.1	32.4	-6.0	33.6	-7.2	35.3
			-3.8	21.7	1.7	1.3	-5.0	22.6	-5.9	23.3	-6.9	24.1	-8.1	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	6.0	5.8	7.2	9.3	11.7	14.5	16.5	16.3	15.0	12.5	9.1	6.8		
	DBStd	4.59	2.87	2.72	2.71	2.63	2.45	2.36	2.16	1.86	2.20	2.57	2.51	2.82		
	HDD10.0	543	124	120	93	42	9	1	0	0	0	7	45	102		
	HDD18.3	2726	381	352	346	269	206	117	66	69	103	183	277	356		
	CDD10.0	880	1	2	6	22	62	137	201	194	150	83	18	4		
	CDD18.3	19	0	0	0	0	0	3	8	5	3	1	0	0		
	CDH23.3	97	0	0	0	0	2	19	44	22	8	2	0	0		
	CDH26.7	17	0	0	0	0	1	3	10	3	0	0	0	0		
(14) Wind	WSAvg	4.9	5.9	5.6	5.5	4.6	4.7	4.4	4.3	4.2	4.4	4.7	5.1	5.8		
(15) Precipitation	PrecAvg	901	101	76	61	60	52	49	49	59	61	101	114	119		
	PrecMax	1159	182	162	157	156	115	111	103	151	121	216	286	254		
	PrecMin	641	18	12	12	7	10	12	15	10	9	27	32	14		
	PrecStd	129	47	43	32	38	27	28	24	35	28	48	57	56		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	14.1	16.7	20.7	22.2	25.8	27.9	26.2	24.6	21.6	16.0	13.4		
		MCWB	11.4	10.1	11.8	14.4	16.3	19.2	20.4	19.3	18.6	17.0	14.6	12.0		
	2%	DB	11.7	11.6	14.0	17.4	19.6	22.4	24.0	22.6	21.5	18.6	14.5	12.6		
		MCWB	10.9	9.8	10.8	12.7	15.1	17.2	18.0	17.9	17.1	15.6	13.0	11.5		
	5%	DB	10.9	10.5	12.2	15.2	17.7	20.4	22.0	21.0	19.7	17.0	13.5	11.7		
		MCWB	10.1	9.3	10.1	11.6	14.1	16.1	17.2	17.0	16.6	14.9	12.2	10.6		
	10%	DB	10.0	9.7	11.0	13.5	16.0	18.8	20.5	19.6	18.4	16.0	12.7	10.9		
		MCWB	9.1	8.6	9.3	10.6	13.1	15.4	16.5	16.3	15.8	14.4	11.5	9.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	11.2	12.5	14.8	17.1	20.2	20.8	20.1	19.1	17.5	15.0	12.6		
		MCDB	12.1	12.6	15.1	19.3	20.1	24.6	27.0	24.7	22.6	20.0	15.8	13.0		
	2%	WB	11.1	10.2	11.4	13.2	15.7	18.0	18.7	18.6	17.9	16.2	13.5	11.7		
		MCDB	11.6	11.0	13.3	16.7	18.9	21.5	22.4	21.4	20.4	17.6	14.2	12.3		
	5%	WB	10.2	9.5	10.5	12.0	14.5	16.7	17.8	17.7	17.1	15.4	12.5	11.0		
		MCDB	10.8	10.3	12.0	14.6	17.1	19.4	21.0	19.9	18.9	16.7	13.3	11.6		
	10%	WB	9.3	8.7	9.6	11.0	13.5	15.7	17.0	16.9	16.3	14.7	11.7	9.9		
		MCDB	9.9	9.5	10.8	13.0	15.6	18.2	19.7	18.9	17.9	15.8	12.6	10.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.4	5.1	6.4	6.5	6.9	7.0	6.7	6.3	5.6	4.5	4.1		
		MCDBR	4.2	5.5	7.2	9.2	9.5	10.0	10.0	8.9	8.0	6.7	5.3	4.3		
		MCWBR	4.4	4.3	4.8	5.7	6.0	6.0	5.6	5.1	4.6	4.6	4.5	4.4		
	5% WB	MCDBR	4.3	4.7	6.4	8.6	8.5	9.0	8.9	7.9	7.3	5.8	5.1	4.4		
		MCWBR	4.7	4.2	4.8	5.6	5.7	5.9	5.4	5.1	4.7	4.5	4.8	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.344	0.376	0.389	0.394	0.390	0.384	0.379	0.368	0.357	0.332	0.322		
	taud		2.420	2.409	2.325	2.313	2.347	2.387	2.405	2.440	2.452	2.466	2.452	2.388		
	Ebn at Noon		714	795	826	856	865	870	869	858	829	771	704	661		
	Edh at Noon		65	82	105	118	119	115	111	103	92	77	62	58		
(44) All-Sky Solar Radiation	RadAvg		0.88	1.60	2.81	4.46	5.28	5.87	5.65	4.68	3.57	2.00	1.07	0.71		
	RadStd		0.07	0.19	0.30	0.38	0.39	0.56	0.36	0.32	0.23	0.16	0.13	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page